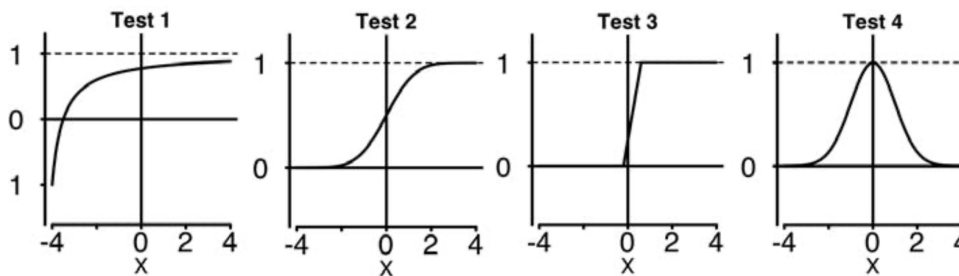


Quiz #3

- Suppose X is a continuous random variable.
 - What is $P(a \leq X \leq a)$?
 - What is $P(X = 0)$?
 - Does $P(X = 2) = 0$ mean that X never equals 2?
- Which of the following are valid cumulative distribution functions? Why are the others not valid?



- Brrrr. Suppose the mean temperature in Pittsburgh for January was 41°F and the variance was 16 °F². The conversion formula between Celsius and Fahrenheit is

$$C = \frac{5}{9}(F - 32).$$

- What was the standard deviation of the temperature in °F?
 - What was the mean temperature in °C?
 - What was the standard deviation of the temperature in °C?
- Suppose exam scores X are approximately normal with mean 70 and standard deviation 4. In words, explain what each line of code calculates. You do not need to compute the numerical values.

```
from scipy.stats import norm
```

```
p1 = norm.cdf(74, loc=70, scale=4)
```

```
p2 = 1 - norm.cdf(74, loc=70, scale=4)
```

```
p3 = norm.cdf(74, loc=70, scale=4) - norm.cdf(66, loc=70, scale=4)
```